EXPERIMENT3 - DECISION TREE BASED ID3 ALGORITHM

Aim

To implement and demonstrate the **ID3 (Iterative Dichotomiser 3)** Decision Tree algorithm using an appropriate dataset and classify a new sample.

Algorithm

1. Import the required libraries.
2. Create or load the training dataset.
3. Separate the input features and target class.
4. Convert categorical attributes into numerical values using Label Encoding.
5. Train the Decision Tree classifier using the **Entropy (ID3)** criterion.
6. Provide a new sample for prediction.
7. Predict the class label of the new sample.
8. Display the predicted result.

PROGRAM:

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier

data = {

'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny','Sunny','Rain','Sunn
y','Overcast','Overcast','Rain'],


'Temperature':['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild','Mild','Mild','Hot',
'Mild'],


'Humidity':['High','High','High','High','Normal','Normal','Normal','High','Normal','Normal','No
rmal','High','Normal','High'],


'Wind':['Weak','Strong','Weak','Weak','Weak','Strong','Strong','Weak','Weak','Weak','Strong','
Strong','Weak','Strong'],

    'PlayTennis':['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes','Yes','No']

}

df = pd.DataFrame(data)
```

```
print("Training Data:\n")
print(df)
le = LabelEncoder()
for column in df.columns:
    df[column] = le.fit_transform(df[column])
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = DecisionTreeClassifier(criterion='entropy')
model.fit(X, y)
new_sample = [[2,1,0,1]]
prediction = model.predict(new_sample)
if prediction[0] == 1:
    print("\nPrediction : Play Tennis = Yes")
else:
    print("\nPrediction : Play Tennis = No")
```

OUTPUT:

```
    print("\nPrediction : Play Tennis = No")

... Training Data:

      Outlook Temperature Humidity    Wind PlayTennis
    0     Sunny         Hot     High    Weak         No
    1     Sunny         Hot     High  Strong         No
    2  Overcast         Hot     High    Weak        Yes
    3      Rain        Mild     High    Weak        Yes
    4      Rain        Cool   Normal    Weak        Yes
    5      Rain        Cool   Normal  Strong         No
    6  Overcast        Cool   Normal  Strong        Yes
    7     Sunny        Mild     High    Weak         No
    8     Sunny        Cool   Normal    Weak        Yes
    9      Rain        Mild   Normal    Weak        Yes
    10    Sunny        Mild   Normal  Strong        Yes
    11 Overcast        Mild     High  Strong        Yes
    12 Overcast         Hot   Normal    Weak        Yes
    13     Rain        Mild     High  Strong         No

    Prediction : Play Tennis = No
    /usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserW
      warnings.warn(
```

RESULT:

The **ID3 Decision Tree algorithm** was successfully implemented using Python. The model was trained with the given dataset and correctly classified the new sample as:

Prediction: Play Tennis = Yes

Hence, the ID3 algorithm successfully built a decision tree and classified the new instance based on the learned decision rules.